

Docker & Containers for QA

Images, containers, Dockerfiles, volumes, ports and why containers help reproduce test environments.

Image vs container

An image is an immutable package with an application and its dependencies. A container is a running instance of that image, isolated from the rest of the system.

Dockerfile and execution

A Dockerfile describes how to build an image. Ports, volumes, environment variables and networks control communication and data persistence.

Use in QA

Containers help create reproducible dependencies, temporary databases, mocks, services and local environments close to the pipeline.

Considerations

Different versions, persisted data, networking, resources and secrets can produce different behavior. Record the image/tag used in evidence when relevant.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

How QA applies it

Start from the behavior or risk to validate, prepare data and environment, execute the technique or tool reproducibly and capture evidence that helps the team understand the result.

QA tip

If a bug occurs only in one environment, comparing container image, configuration and variables can reveal important differences.

Official resources

Site oficial: <https://www.docker.com/>